Highlights

Robustness indices reward poor nominal schedules: a measurement audit of dynamic job-shop scheduling under disruption

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- The conventional schedule robustness index is proved to be a strictly decreasing function of nominal deviation at fixed realised performance.
- It correlates with nominal deviation in all nine disruption regimes, at $\rho_s = -0.95$ to -0.98 where ten methods are compared: worse nominal schedules score as more robust.
- Ranking by the index contradicts ranking by realised quality in all nine regimes, costing up to 49.3 percentage points of deviation from the optimum.
- Realised quality against a certified optimum, plus a CVaR_{90} tail, restores cross-method comparability without new experiments.
- Under the corrected metric a GNN–PPO policy matches the best comparator in every regime and ranks first in five, including all arrival regimes.

Robustness indices reward poor nominal schedules: a measurement audit of dynamic job-shop scheduling under disruption

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Abstract

Deep reinforcement learning (DRL) is increasingly proposed for dynamic job-shop scheduling on the grounds that learned dispatching policies are more *robust* to disruption than offline optimisation. That claim is almost always supported by a robustness index of the form $\mathrm{RI} = (\mathbb{E}[C^{\mathrm{pert}}_{\max}] - C^{\mathrm{nom}}_{\max})/C^{\mathrm{nom}}_{\max}$, which normalises the degradation a method suffers by its own undisrupted makespan. We show, analytically and empirically, that this self-referential normalisation makes the index unusable for comparing methods. Writing r and q for the relative deviations of the nominal and the realised makespan from a certified optimum, we prove that $\mathrm{RI} = (q-r)/(1+r)$, so that at equal realised performance RI is strictly decreasing in r: the worse a method's nominal schedule, the more robust it appears. In a corpus of 4,500 disrupted executions spanning 9 pre-registered regimes across three disruption classes, RI is rank-correlated with nominal deviation in every regime, at ρ_s between -0.985 and -0.948 ($p \le 3\times10^{-5}$) across the 6 regimes that compare ten methods, and the ordering it induces disagrees with the ordering by realised quality in all 9 (Kendall τ negative throughout). Selecting a scheduler by RI rather than by realised quality costs between 15.8 and 49.3 percentage points of deviation from the optimum (median 25.7). We propose realised quality — the deviation of the executed makespan from the certified optimum — together with its CVaR_{90} tail, and a three-comparator reporting discipline that separates the mean rule, the best rule per family and the per-instance oracle. Re-measured this way, a heterogeneous graph-neural-network PPO policy is statistically indistinguishable from the best comparator in all 9 regimes and ranks first in 5, including all three dynamic-arrival regimes — a conclusion the conventional index inverts.

1. Introduction

Production schedules are computed under assumptions that the shop floor does not honour. Machines break down, processing times realise away from their estimates, and jobs arrive after the schedule has been fixed. The scheduling literature has responded with two broad strategies: compute a schedule offline and repair it as events unfold, or learn a dispatching policy that decides online and therefore never commits to a sequence it must later abandon (Ouelhadj and Petrovic, 2009; Aytug, Lawley, McKay, Mohan and Uzsoy, 2005). The second strategy has attracted intense recent interest, with graph neural networks supplying a size-agnostic state representation and policy-gradient methods supplying the learning signal (Zhang, Song, Cao, Zhang, Tan and Xu, 2020; Park, Chun, Kim, Kim and Park, 2021; Song, Chen, Li and Cao, 2023; Bengio, Lodi and Prouvost, 2021).

The empirical case for learned policies rests on a robustness claim. A learned policy is said to degrade less under disruption than an offline optimum does, and this is quantified by an index that compares a method's disrupted makespan to its own undisrupted makespan:

$$\mathrm{RI} \;=\; \frac{\mathbb{E}[C^{\mathrm{pert}}_{\max}] - C^{\mathrm{nom}}_{\max}}{C^{\mathrm{nom}}_{\max}}. \tag{1}$$

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Variants of (1) appear throughout the robust-scheduling literature (Leon, Wu and Storer, 1994; Mehta and Uzsoy, 1998; Goren and Sabuncuoglu, 2008; Sabuncuoglu and Bayiz, 2000) and have been carried into the DRL-scheduling literature largely unexamined (Smit, Zhou and Bekker, 2025; Huang, Ye and Yang, 2025; Xu, Mei, Zhang and Zhang, 2025). The index has an appealing reading: it is dimensionless, it is comparable across instances of different size, and a value near zero says that disruption cost the method little.

This paper argues that (1) cannot bear the comparative weight placed on it, and that the problem is structural rather than a matter of estimation noise. The denominator is the method's own nominal makespan. A method that produces a poor nominal schedule is therefore graded against a lenient reference, and one that produces a near-optimal schedule is graded against a demanding one. Section 4 makes this exact: with r the relative deviation of the nominal makespan from a certified optimum and q the relative deviation of the realised makespan from the same optimum, $\mathrm{RI} = (q-r)/(1+r)$, whose partial derivative in r is strictly negative. Two methods that deliver *identical* executed schedules are ranked by RI in reverse order of the quality of the plans they started from.

The consequence is not hypothetical. We re-analyse a pre-registered corpus of 4,500 disrupted executions covering three disruption classes at three severities each — machine breakdowns, stochastic processing times and dynamic job arrivals — for nine priority dispatching rules and a heterogeneous GNN–PPO policy, together with 43 nominal instances with proven optima. In every one of the 9 regimes, RI is negatively rank-correlated with how bad the nominal schedule was — almost perfectly so in the 6 regimes that compare ten methods — and the method it nominates as most robust is never the method that actually finishes closest to the optimum. Under breakdowns and stochastic processing times the index crowns the critical-ratio rule, whose nominal schedules sit some 220% above the optimum; under dynamic arrivals it crowns least-work-remaining, at 187%. Both are among the worst performers once the executed schedule is measured against a fixed reference.

Contributions. This paper makes four.

(i) **An analytic characterisation of the failure mode.** Proposition 1 decomposes RI into nominal and realised deviation and shows the monotone dependence on the former; Corollary 2 identifies negative RI — routinely reported as strong robustness — as the signature of a nominal schedule so poor that perturbation improves it in expectation.

(ii) **A quantification of the resulting decision error.** Across 9 regimes, ranking by RI and ranking by realised quality disagree without exception, and acting on the former costs 15.8–49.3 percentage points of deviation from the optimum.

(iii) **A corrected measurement protocol.** Realised quality against a certified optimum, reported jointly with a CVaR_{90} tail and against three explicitly separated dispatching comparators, restores cross-method comparability. The protocol requires no additional experiments: it is a re-measurement of runs already performed.

(iv) **A re-evaluation of the learned policy under that protocol.** The GNN–PPO policy is not significantly worse than the best-ranked comparator in any of the 9 regimes at the Nemenyi critical difference and ranks first in 5, including all three dynamic-arrival regimes — while the conventional index places it near the bottom. We are careful throughout to read a failure to reject as a tie rather than as a win.

Section 2 situates the argument. Section 3 fixes the problem and the disruption model. Section 4 develops the measurement framework. Section 5 describes the experimental design and the pre-registration that governs it. Section 6 reports the audit. Section 7 discusses implications and limitations, and Section 8 concludes.

2. Related work

2.1. Learning to schedule

Dispatching rules remain the operational default in dynamic shops because they are cheap, interpretable and require no state beyond what the shop already tracks (Haupt, 1989; Holthaus and Rajendran, 1997). Their weakness is that no single rule dominates: performance depends on shop load, due-date tightness

and the disruption regime. Learning a dispatching policy is an attempt to obtain the online character of a rule with the instance-adaptivity of search. Zhang et al. (2020) framed job-shop scheduling as sequential decision making over a disjunctive graph and learned a policy with proximal policy optimisation (Schulman, Wolski, Dhariwal, Radford and Klimov, 2017); Park et al. (2021) and Song et al. (2023) extended the idea to richer graph encodings and to the flexible variant. Recent surveys (Kayhan and Yildiz, 2023; Smit et al., 2025; Huang et al., 2025; Xu et al., 2025) document rapid growth in the area and, notably, converge on the observation that evaluation practice is heterogeneous enough to make cross-paper comparison difficult.

2.2. Robustness measurement in scheduling

Robust scheduling distinguishes *robustness* — retained solution quality under realisation — from *stability*, the degree to which the executed schedule resembles the planned one (Leon et al., 1994; Goren and Sabuncuoglu, 2008). Both are usually measured relative to the method's own plan. Leon et al. (1994) define robustness through the expected makespan increase relative to the deterministic schedule; Mehta and Uzsoy (1998) measure predictability similarly; Sabuncuoglu and Bayiz (2000) and Aytug et al. (2005) survey the reactive-scheduling variants of the same construction. Herroelen and Leus (2005) note in the project-scheduling context that such measures conflate the quality of the baseline with its resilience, but the observation has not propagated into machine-scheduling evaluation practice, and in particular not into the DRL-scheduling literature, where the index is typically inherited without an examination of what its denominator does.

Our contribution here is to make that conflation precise and to show that it is not a second-order concern: in the regimes we study it is the dominant term, and it inverts the induced ordering.

2.3. Evaluation hygiene in learning-based methods

The reproducibility literature in reinforcement learning has repeatedly found that reported gains are sensitive to seeds, to the choice of baseline and to the statistical procedure (Henderson, Islam, Bachman, Pineau, Precup and Meger, 2018; Agarwal, Schwarzer, Castro, Courville and Bellemare, 2021). The remedies proposed there — pre-registration of hypotheses, explicit comparator definitions, distribution-free tests over instance sets, and reporting of tails rather than means alone (Demšar, 2006; García and Herrera, 2008; Munafò, Nosek, Bishop, Button, Chambers, Percie du Sert, Simonsohn, Wagenmakers, Ware and Ioannidis, 2017; Kerr, 1998) — transfer directly to scheduling, where the instance set plays the role of the task distribution. This paper applies them, and adds one scheduling-specific requirement: the reference against which quality is measured must be external to the method being graded.

3. Problem setting

3.1. The dynamic flexible job-shop problem

An instance comprises a set of jobs $\mathcal{J} = \{J_1, \dots, J_n\}$ and machines $\mathcal{M} = \{M_1, \dots, M_m\}$. Job J_i is a chain of operations $O_{i1} \prec O_{i2} \prec \cdots \prec O_{in_i}$; operation O_{ij} may be processed on any machine in an eligibility set $\mathcal{M}_{ij} \subseteq \mathcal{M}$, taking p_{ijk} time units on $M_k \in \mathcal{M}_{ij}$. In the classical job shop $|\mathcal{M}_{ij}| = 1$ for all (i, j). Machines process one operation at a time and operations are non-preemptable once started. A solution assigns each operation a machine and a start time consistent with precedence and machine capacity; the objective is the makespan $C_{\max} = \max_i C_{in_i}$.

The dynamic variant lets the instance change during execution. We write ξ for a realisation of the disruption process and $C^{\mathrm{pert}}_{\max}(\xi)$ for the makespan actually achieved when a method's decisions are executed against ξ.

3.2. Disruption classes

Three disruption classes are modelled, each at three severities, giving the 9 regimes of the experimental grid. The parameterisations were fixed before any result was observed (Section 5.5).

Machine breakdowns. Each machine alternates between available and under-repair states calibrated to a target availability $A_k \in \{0.95, 0.90, 0.80\}$. Times to failure are exponential and times to repair log-normal with shape $\sigma_{\mathrm{TTR}} = 0.5$ and a mean-time-to-repair multiplier of 2.0. An operation in progress when its machine fails resumes on repair.

Stochastic processing times. Realised durations are drawn as $\tilde{p}_{ijk} = p_{ijk} \cdot \varepsilon$ with $\varepsilon \sim$ LogNormal calibrated to unit mean and coefficient of variation $\sigma \in \{0.10, 0.25, 0.50\}$, so that the expected instance is unchanged and only its dispersion varies.

Dynamic job arrivals. New jobs arrive as a Poisson process whose rate is calibrated to a target machine utilisation $\rho \in \{0.70, 0.85, 0.95\}$, up to a horizon after which no further arrivals occur.

3.3. Execution semantics

The distinction between methods that decide once and methods that decide continuously must be made explicit, because it determines what a disrupted makespan even means.

An *offline* method (constraint programming, a genetic algorithm) produces a complete schedule against the nominal instance. Under disruption that schedule is executed with right-shift repair: the machine sequence is preserved and start times are pushed forward as needed to respect the realised durations, availabilities and release dates. An *online* method (a dispatching rule, a learned policy) is invoked at each decision point against the state as it actually is, and therefore never holds a sequence it must repair.

This asymmetry is intrinsic to the comparison and we do not attempt to remove it. We do, however, decline to let it hide inside a metric: the measurement framework of Section 4 grades both families against the same external reference.

Not every dispatching rule survives the arrival regimes. 5 of the nine are excluded there, for two distinct reasons. EDD and CR require job due dates, and the due dates of jobs that did not exist when the instance was created are not defined by the instance; synthesising them online is a modelling decision, not an implementation detail. SPT, LPT and MWKR require per-operation duration or remaining-work quantities indexed against the original operation set, which the implementation does not maintain for dynamically arriving jobs. Four rules (FIFO, MOR, LOR, LWKR) plus the learned policy are therefore compared under arrivals, against ten methods elsewhere.

This restriction matters for how the arrival results should be read, and we flag it here rather than in a footnote: MWKR, which outranks the learned policy in five of the six non-arrival regimes, is among the rules excluded. The arrival comparison is therefore against a weaker field than the breakdown and stochastic-time comparisons, and the policy's first-place finish there is a first place among four rules, not among nine. Section 7.4 returns to this. Removing the restriction requires tracking each job's true arrival time and recomputing rule state against a growing operation set — a change of state representation we did not attempt, and disclose rather than resolve.

4. A measurement framework

4.1. The conventional index and what its denominator does

Definition 1 (Robustness index). For a method producing nominal makespan $C^{\text{nom}}_{\max}$ and expected disrupted makespan $\mathbb{E}[C^{\text{pert}}_{\max}]$ on the same instance,

$$\text{RI} = \frac{\mathbb{E}[C^{\text{pert}}_{\max}] - C^{\text{nom}}_{\max}}{C^{\text{nom}}_{\max}}.$$

Let $C^{\star}_{\max}$ denote a certified optimum for the nominal instance and define the two *externally referenced* deviations

$$r = \frac{C^{\text{nom}}_{\max} - C^{\star}_{\max}}{C^{\star}_{\max}}, \qquad q = \frac{\mathbb{E}[C^{\text{pert}}_{\max}] - C^{\star}_{\max}}{C^{\star}_{\max}}. \tag{2}$$

Here r is the familiar relative percentage deviation of the plan and q is what we will call *realised quality*: how far the schedule that was actually executed finished from the best schedule that existed for the nominal instance. Both are measured against a reference that no method can influence.

Proposition 1. *With r and q as in (2) and $C^{\star}_{\max} > 0$,*

$$\text{RI} = \frac{q - r}{1 + r}, \qquad \textit{and} \qquad \left.\frac{\partial \text{RI}}{\partial r}\right|_{q} = -\frac{1 + q}{(1 + r)^2} < 0. \tag{3}$$

PROOF. Substituting $C^{\mathrm{nom}}_{\max} = C^{\star}_{\max}(1+r)$ and $\mathbb{E}[C^{\mathrm{pert}}_{\max}] = C^{\star}_{\max}(1+q)$ into Definition 1 gives $\mathrm{RI} = \big(C^{\star}_{\max}(1+q) - C^{\star}_{\max}(1+r)\big)/C^{\star}_{\max}(1+r) = (q-r)/(1+r)$. Differentiating in r at fixed q, $\partial\mathrm{RI}/\partial r = \big[-(1+r)-(q-r)\big]/(1+r)^2 = -(1+q)/(1+r)^2$, which is negative whenever $q > -1$, i.e. whenever the disrupted makespan is positive. □

Corollary 1 (Rank inversion at equal performance)**.** *Two methods with identical realised quality q but nominal deviations $r_1 < r_2$ satisfy $\mathrm{RI}_1 > \mathrm{RI}_2$. The method with the worse plan receives the better robustness score.*

Corollary 2 (Negative index)**.** $\mathrm{RI} < 0$ *if and only if $q < r$: the expected disrupted makespan is* below *the nominal one. This can only occur when the nominal schedule was poor enough that the disruption process, by reordering work it had sequenced badly, improves it in expectation.*

Corollary 2 deserves emphasis because negative robustness indices are reported in the literature as evidence of exceptional resilience. Under Definition 1 they are instead a diagnostic of a bad baseline. A method cannot benefit from machine failures; it can only have been scheduling so inefficiently that the failures were not the binding constraint.

Proposition 1 is an identity, not an approximation. Evaluated over the 150 (method × instance × regime) cells of our corpus, the maximum absolute discrepancy between RI as computed from raw makespans and $(q-r)/(1+r)$ as computed from the stored deviations is 9.4×10^{-6}; computed from raw makespans on both sides the discrepancy falls to floating-point noise, confirming that the residual is rounding in the recorded deviation columns.

4.2. Realised quality

We take q of (2) as the primary performance measure under disruption and call it realised quality, RQ. It has three properties the conventional index lacks. It is referenced externally, so a method cannot improve its score by planning badly. It is comparable across methods, because all methods on an instance share the denominator $C^{\star}_{\max}$. And it is comparable to the nominal deviation r on the same scale, so the cost of disruption is read as the difference $q - r$ rather than being folded into a ratio.

It also has a limitation worth stating. RQ measures the executed schedule against the optimum *of the nominal instance*, which under dynamic arrivals is not the optimum of the instance that was actually executed. The quantity is therefore a lower-bound-referenced deviation rather than a true optimality gap in the arrival regimes; it remains comparable across methods, which is what the comparison requires, but its absolute magnitude there should not be read as an optimality gap. We report arrival regimes separately throughout for this reason.

4.3. Tail risk

A mean conceals the executions a shop actually fears. We therefore report, alongside $\mathbb{E}[\mathrm{RQ}]$, the conditional value at risk at the 90th percentile,

$$\mathrm{CVaR}_{90}(\mathrm{RQ}) = \mathbb{E}\big[\mathrm{RQ} \mid \mathrm{RQ} \geq \mathrm{VaR}_{90}(\mathrm{RQ})\big], \tag{4}$$

the mean realised quality over those realisations at or above the 90th percentile (Rockafellar and Uryasev, 2000; Artzner, Delbaen, Eber and Heath, 1999). With 30 seeds per (instance, method) this averages the worst 3 realisations, so it is a coarse tail estimate and we read it as a direction, not a precise quantile. Two methods with equal expected performance and different tail behaviour are operationally different, and a scheduler chosen for a shop that cannot absorb a late finish should be chosen on the tail.

4.4. Comparator discipline

Claims of the form "the policy beats the best dispatching rule" are underdetermined until *best* is defined, and the available definitions do not agree. We report all three, always separately and always labelled.

$\mathrm{PDR}_{\mathrm{global}}$ the mean deviation over all nine rules: what a practitioner obtains who has not tuned rule selection.

$\mathrm{PDR}_{\mathrm{family}}$ the mean, over the five behavioural families (duration-based, work-remaining, operations-remaining, arrival-based, due-date-based), of the best rule *within* that family on that instance. This is an intermediate construction: it is stronger than any single deployable rule because the within-family choice is made per instance, but weaker than the oracle below because it cannot cross family boundaries.

$\mathrm{PDR}_{\mathrm{oracle}}$ the minimum deviation per instance: attainable only with knowledge of the answer, and therefore an upper bound on rule-based performance rather than a deployable baseline.

Section 6.5 shows these three yield opposite verdicts on the same data, which is precisely why a paper that reports one without naming it has not reported a result.

5. Experimental design

5.1. Instances and reference values

The nominal corpus is the 43 standard job-shop instances bundled in the OR-Library — `ft06`, `ft10`, `ft20` (Fisher and Thompson, 1963) and `la01`–`la40` (Lawrence, 1984). Every instance in this set has a proven optimum, verified as such in the accompanying reference table, so the deviation r is a true optimality gap and not a best-known-bound approximation. The disruption grid uses `ft06` (6×6) and `la01` (10×5), the pair against which the disruption machinery was validated.

5.2. Methods

Nine priority dispatching rules span five behavioural families: shortest and longest processing time (SPT, LPT); most and least work remaining (MWKR, LWKR); most and least operations remaining (MOR, LOR); first in first out (FIFO); and earliest due date and critical ratio (EDD, CR). A constraint-programming model solved with CP-SAT (Perron and Furnon, 2024) provides the exact reference and a generational genetic algorithm the metaheuristic reference. The learned method is a heterogeneous graph neural network encoder in the GIN family (Xu, Hu, Leskovec and Jegelka, 2019) over a graph with distinct precedence, machine-sharing and eligibility edge types, trained with maskable PPO (Schulman et al., 2017; Raffin, Hill, Gleave, Kanervisto, Ernestus and Dormann, 2021) under a domain-randomised curriculum (Tobin, Fong, Ray, Schneider, Zaremba and Abbeel, 2017). Hyper-parameters were fixed by a search with Optuna (Akiba, Sano, Yanase, Ohta and Koyama, 2019) conducted before the confirmatory campaign and reported separately; the campaign analysed here uses the resulting configuration unchanged.

5.3. Grid and replication

Each of the 9 regimes of Section 3.2 is executed for 30 independent disruption seeds on each of the two instances, giving 30×2 paired blocks per regime and 4,500 disrupted executions in total. Ten methods are compared in the six breakdown and stochastic-time regimes and five in the three arrival regimes, for the reason given in Section 3.3.

Nominal reference convention. Exactly one deterministic run per (instance, method) enters the nominal reference. Each baseline contributes its single deterministic schedule; the learned policy contributes its deterministic rollout, and the four additional stochastic rollouts recorded for it are reported descriptively and never enter a test, per the pre-registration. This keeps the unit of analysis identical across methods.

5.4. Statistical protocol

Within a regime, methods are compared by a Friedman test over blocks defined by (instance, seed), followed — on rejection — by a Nemenyi post-hoc with critical difference $\mathrm{CD} = q_\alpha \sqrt{k(k+1)/6N}$ for k methods and N blocks (Demšar, 2006; García and Herrera, 2008). Paired two-method comparisons use the Wilcoxon signed-rank test with the rank-biserial correlation as effect size. Association between metrics uses Spearman's ρ_s and, for ordering agreement, Kendall's τ. All tests are two-sided at $\alpha = 0.05$. Because the audit reports one Friedman test per regime and these regimes are pre-registered as separate confirmatory families rather than as a single pooled hypothesis, no cross-regime multiplicity correction is applied; the per-regime p-values are reported in full so that any preferred correction can be applied by the reader.

Table 1
Rank correlation between nominal deviation and the conventional robustness index, by regime. k is the number of methods compared. A strongly negative ρ_s means that methods with worse nominal schedules receive better robustness scores.

Disruption class	Severity	k	ρ_s	p
Breakdowns	$A_k = 0.95$	10	-0.985	2.3×10^{-7}
	$A_k = 0.90$	10	-0.948	2.9×10^{-5}
	$A_k = 0.80$	10	-0.948	2.9×10^{-5}
Stochastic times	$\sigma = 0.10$	10	-0.948	2.9×10^{-5}
	$\sigma = 0.25$	10	-0.985	2.3×10^{-7}
	$\sigma = 0.50$	10	-0.948	2.9×10^{-5}
Dynamic arrivals	$\rho = 0.70$	5	-0.800	0.104
	$\rho = 0.85$	5	-0.800	0.104
	$\rho = 0.95$	5	-0.800	0.104

5.5. Pre-registration and reproducibility

The hypotheses, the instance set, the disruption grid, the statistical tests and the accept/reject rules were frozen in a dated pre-registration document before any policy was trained or any disrupted run executed, following the recommendations of Munafò et al. (2017) and the specific concern about post-hoc hypothesis formation raised by Kerr (1998). The analysis reported here is a *re-measurement* of the pre-registered corpus: no run was added, removed or re-executed, and the raw per-run records are unchanged. The re-measurement is itself post-hoc with respect to that freeze — it was motivated by an audit of the metric, not registered in advance — and is labelled as such throughout. What it does not do is alter any datum on which the original verdicts were computed.

Appendix A sets out the research programme from which the corpus arises.

6. Results

6.1. The robustness index is determined by nominal quality

Table 1 reports, for each regime, the Spearman rank correlation between a method's nominal deviation and its robustness index.

The correlation is negative in all 9 regimes. In the 6 breakdown and stochastic-time regimes, where ten methods are compared, ρ_s ranges from -0.985 to -0.948 (median -0.948) with $p \leq 3 \times 10^{-5}$ — an ordering that is very nearly deterministic. The 3 arrival regimes give $\rho_s = -0.800$ throughout, but with only five methods a Spearman test over $k = 5$ points has little power and we report those three descriptively rather than as confirmation. Figure 1 shows the underlying relation: the index is, to a good approximation, an affine decreasing function of the nominal deviation, exactly as Proposition 1 predicts.

97 of the 150 (method × instance × regime) cells in the corpus — 65% — have RI < 0. By Corollary 2 each of these is a case in which the expected disrupted makespan is below the nominal one, and we verify that the equivalence holds exactly: in 100% of cells, sign(RI) agrees with sign$(q - r)$. The critical-ratio rule on `ft06` under $A_k = 0.90$ is representative — its nominal makespan of 171 falls to a mean of 94.1 under breakdowns. The rule computes a criticality ratio from due dates that were synthesised once against nominal durations and are never recomputed as the realised clock advances, so under disruption its ordering decisions become progressively less related to the state it is ordering; the resulting near-random reordering improves on a sequence that was badly chosen to begin with. This is regression toward the mean, and the index records it as resilience.

6.2. Rank reversal and the cost of acting on it

If the index merely added noise, it would still be usable in expectation. It does not: it induces an ordering that is systematically opposed to the ordering by realised quality.

Kendall's τ is negative in all 9 regimes (Table 2, range -0.60 to -0.29), and the two criteria never select the same method. The per-regime τ tests are individually significant in only a minority of regimes — with at most ten methods each they have little power — but the sign is negative without exception, and it is the

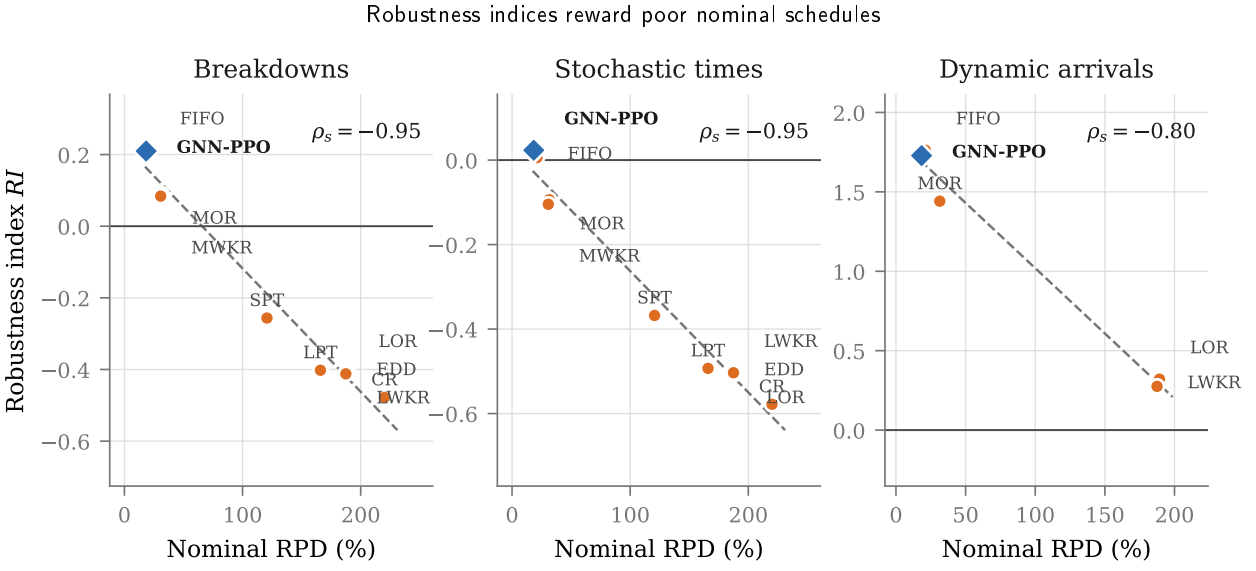


Figure 1: The robustness index against nominal deviation, aggregated by disruption class. Each point is one method. The dashed line is an ordinary least-squares fit shown for orientation only; the reported statistic is the rank correlation. Methods whose nominal schedules sit furthest from the optimum (CR, LWKR, LOR, EDD) obtain the most negative — conventionally, the most robust — indices.

Table 2
Disagreement between the two orderings. τ is Kendall's rank correlation between the ordering by RI and the ordering by realised quality. The last three columns give the realised quality of the method the index nominates, that of the method that is actually best, and the penalty incurred by following the index.

Disruption class	Severity	τ	Selected method		Realised quality (%)		
			by RI	by RQ	of former	of latter	penalty
Breakdowns	$A_k = 0.95$	−0.600	CR	MWKR	46.9	23.6	+23.3
	$A_k = 0.90$	−0.422	CR	MWKR	63.3	34.8	+28.5
	$A_k = 0.80$	−0.511	CR	**GNN–PPO**	90.6	64.9	+25.7
Stochastic times	$\sigma = 0.10$	−0.289	CR	MWKR	30.6	12.3	+18.4
	$\sigma = 0.25$	−0.378	CR	MWKR	33.7	13.7	+20.0
	$\sigma = 0.50$	−0.511	CR	MWKR	40.8	24.9	+15.8
Dynamic arrivals	$\rho = 0.70$	−0.400	LWKR	**GNN–PPO**	268.0	230.4	+37.6
	$\rho = 0.85$	−0.600	LWKR	**GNN–PPO**	268.1	219.2	+48.9
	$\rho = 0.95$	−0.400	LWKR	**GNN–PPO**	261.1	211.7	+49.3

consistency of that sign across nine independent regimes, not any single p-value, that carries the claim. The index selects the critical-ratio rule in all 6 breakdown and stochastic-time regimes and least-work-remaining in all 3 arrival regimes. Realised quality selects most-work-remaining or the learned policy. The penalty for following the index — the difference in realised quality between the method it nominates and the method that is actually best — ranges from 15.8 to 49.3 percentage points, with a median of 25.7. Figure 2 shows the reversal directly.

6.3. Realised quality and tail risk

Table 3 gives the full metric panel. Reading the columns against each other is instructive: the ordering of the RI column is close to the reverse of the ordering of the RQ column, and the nominal-deviation column explains the former almost entirely.

Under breakdowns the learned policy attains RQ = 41.8% with $\text{CVaR}_{90} = 85.3\%$ — the best tail in the panel — while MWKR attains a marginally better mean (41.7%) with a worse tail (88.0%). The critical-ratio rule, which the conventional index ranks first, realises 67.0% with a 131.1% tail. Under stochastic processing

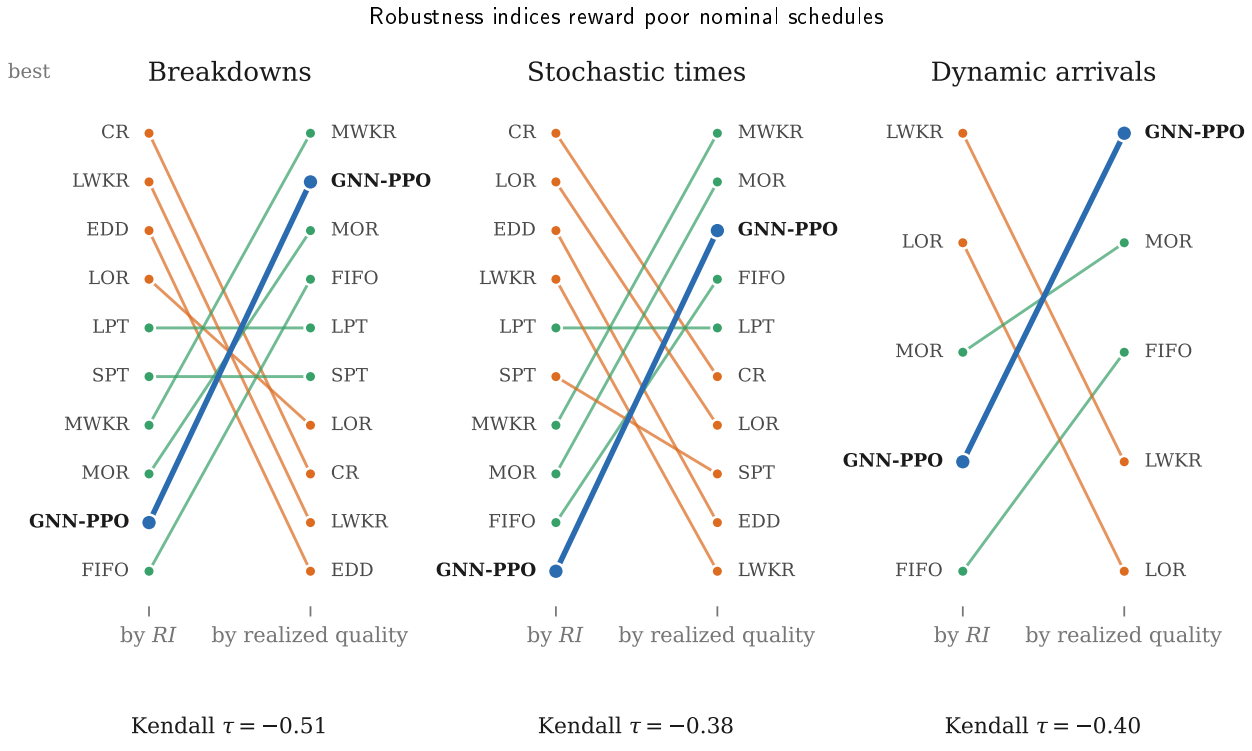


Figure 2: Ordering by the robustness index (left) against ordering by realised quality (right), by disruption class. Lines that fall from left to right are methods the index flatters. The learned policy travels from near-worst to best; the due-date rules travel in the opposite direction.

times MWKR leads on both mean and tail, with the policy third. Under dynamic arrivals the policy leads on the mean and MOR on the tail, the two being close; the absolute magnitudes in this class are large for the reason given in Section 4 and should be read comparatively. Figure 3 displays mean and tail jointly.

6.4. Method separation under the corrected metric

The Friedman null is rejected in every regime with $p \leq 6.7 \times 10^{-28}$ (Table 4), so the methods are distinguishable on realised quality everywhere. The learned policy is the top-ranked method in 5 of 9 regimes and is not significantly worse than the top-ranked method in any of the 9.

Under dynamic arrivals the policy takes first rank in all three regimes, at average ranks of 1.57–1.62 against a critical difference of 0.79. It is not, however, separated from the whole field: MOR trails it by only 0.11–0.20 rank points, well inside the critical difference, so the correct statement is that the policy and MOR are statistically tied at the top while FIFO, LWKR and LOR fall outside it. Figure 4 shows the critical-difference diagrams, in which the clique bar spans exactly that pair. The same reading applies in the other six regimes, where the policy's clique also contains MWKR, MOR and FIFO: nowhere in this corpus does the learned policy separate from every rule.

6.5. Comparator sensitivity

Table 5 evaluates the three comparators of Section 4.4 on the 43-instance nominal corpus.

The verdict on the learned policy depends entirely on which comparator is chosen. Against the mean of the nine rules it wins by 233.4 percentage points; against the best rule per family, by 132.4; against the per-instance oracle, it *loses* by 7.6. All three differences are significant over $N = 43$ paired instances ($p \leq 4.1 \times 10^{-5}$; exact values in Table 5). A paper reporting only the first would claim a decisive win; one reporting only the third, a decisive loss. Both would be describing the same runs. Figure 5 shows the three distributions.

Table 3
Metric panel by disruption class, methods ordered by realised quality. r is nominal deviation from the certified optimum; RI the conventional index; RQ realised quality; CVaR_{90} its worst-decile mean. All percentages. ⋆ marks the learned policy.

Method	r (%)	RI	RQ (%)	CVaR_{90} (%)
Breakdowns				
MWKR	30.7	+0.084	41.7	88.0
GNN–PPO⋆	18.4	+0.210	41.8	85.3
MOR	31.5	+0.090	43.2	85.6
FIFO	21.1	+0.212	46.8	89.6
LPT	165.9	-0.402	57.7	107.4
SPT	120.7	-0.256	64.2	122.8
LOR	189.2	-0.410	65.8	130.3
CR	220.1	-0.478	67.0	131.1
LWKR	187.4	-0.412	68.8	134.6
EDD	187.4	-0.411	69.2	129.7
Stochastic times				
MWKR	30.7	-0.105	17.0	36.0
MOR	31.5	-0.094	19.1	38.0
GNN–PPO⋆	18.4	+0.023	20.0	43.4
FIFO	21.1	+0.006	21.9	49.4
LPT	165.9	-0.493	33.2	59.8
CR	220.1	-0.578	35.0	67.0
LOR	189.2	-0.505	38.6	69.8
SPT	120.7	-0.368	39.6	63.8
EDD	187.4	-0.505	42.2	71.4
LWKR	187.4	-0.503	42.6	74.6
Dynamic arrivals				
GNN–PPO⋆	18.4	+1.728	220.4	275.1
MOR	31.5	+1.441	221.1	273.4
FIFO	21.1	+1.762	234.7	287.1
LWKR	187.4	+0.275	265.7	307.6
LOR	189.2	+0.321	266.4	310.6

6.6. What optimality buys under disruption

A supplementary comparison bears on the motivating claim of the DRL-scheduling literature. On `ft06` under breakdowns at $A_k = 0.90$, CP-SAT and the genetic algorithm both reach the proven optimum nominally ($r = 0$). Executed with right-shift repair over five shared disruption seeds, they realise 42.0% and 41.3% above the optimum respectively — against 44.1% for MOR, 46.8% for MWKR and 48.0% for the learned policy. The paired differences between the policy and either the exact or the metaheuristic method are not significant at this sample size ($p = 0.375$ and $p = 0.250$; one seed produces an exact tie for all three methods, leaving $n = 4$ non-zero differences). The genetic algorithm consumes 30 s per instance and CP-SAT 0.046 s, against 0.095 s for the policy's full rollout.

The reading we take from this is deliberately cautious. Five seeds on one instance in one regime is a descriptive observation, not a test, and we report it as such. What it does suggest is that nominal optimality is largely dissipated by right-shift execution, which is the substantive question the robustness index was introduced to answer and which realised quality answers directly.

7. Discussion

7.1. What the index was measuring

Proposition 1 shows that RI is a normalised difference between realised and nominal quality, and Section 6.1 shows that in practice the nominal term dominates. The index is therefore best understood

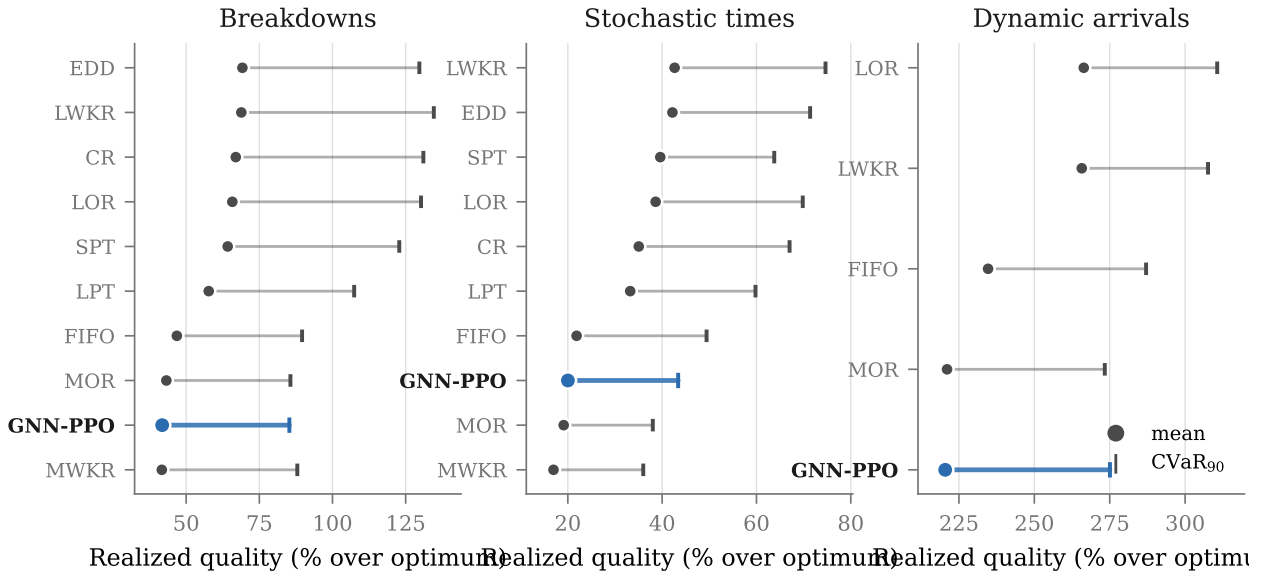


Figure 3: Realised quality by method and disruption class. The circle is the mean, the tick the CVaR_{90}, and the segment between them the tail exposure. Methods are ordered by mean realised quality.

Table 4
Friedman tests on realised quality with Nemenyi post-hoc, by regime. Blocks are (instance, seed) pairs. The last column records whether the learned policy is statistically indistinguishable from the best-ranked method at the critical difference.

Disruption class	Severity	k	N	χ^2	p	CD	Best-ranked	GNN–PPO rank	Tied
Breakdowns	$A_k = 0.95$	10	60	256.0	$< 10^{-8}$	1.75	MWKR	2.83	✓
	$A_k = 0.90$	10	60	252.5	$< 10^{-8}$	1.75	**GNN–PPO**	2.83	✓
	$A_k = 0.80$	10	60	170.4	$< 10^{-8}$	1.75	**GNN–PPO**	3.12	✓
Stochastic times	$\sigma = 0.10$	10	60	289.1	$< 10^{-8}$	1.75	MWKR	3.28	✓
	$\sigma = 0.25$	10	60	267.3	$< 10^{-8}$	1.75	MWKR	3.15	✓
	$\sigma = 0.50$	10	60	150.6	$< 10^{-8}$	1.75	MWKR	3.85	✓
Dynamic arrivals	$\rho = 0.70$	5	60	183.0	$< 10^{-8}$	0.79	**GNN–PPO**	1.62	✓
	$\rho = 0.85$	5	60	200.4	$< 10^{-8}$	0.79	**GNN–PPO**	1.57	✓
	$\rho = 0.95$	5	60	190.1	$< 10^{-8}$	0.79	**GNN–PPO**	1.57	✓

not as a robustness measure that happens to be noisy but as a measure of a different quantity — roughly, how much headroom a method left itself — reported under a name that invites the wrong interpretation. A method with no headroom, because it scheduled well, cannot score well on it.

This does not make the underlying question empty. "How much of its planned quality does a method retain under realisation?" is a real and important question, and the difference $q - r$ answers it on an interpretable scale. What must be abandoned is the normalisation by $C^{\mathrm{nom}}_{\max}$, and with it the practice of comparing RI across methods.

7.2. Implications for reported results

We are not in a position to re-analyse other authors' data, and we make no claim about the correctness of any specific published result. What Proposition 1 does establish is conditional and, we think, consequential: wherever a comparison of methods rests on an index of the form of Definition 1, and wherever the compared methods differ appreciably in nominal quality, the reported ordering is confounded by that difference in a known direction and by a computable amount. Since learned policies and dispatching rules routinely differ by a factor of several in nominal deviation — on the two disruption instances of this corpus, 18.4% for the

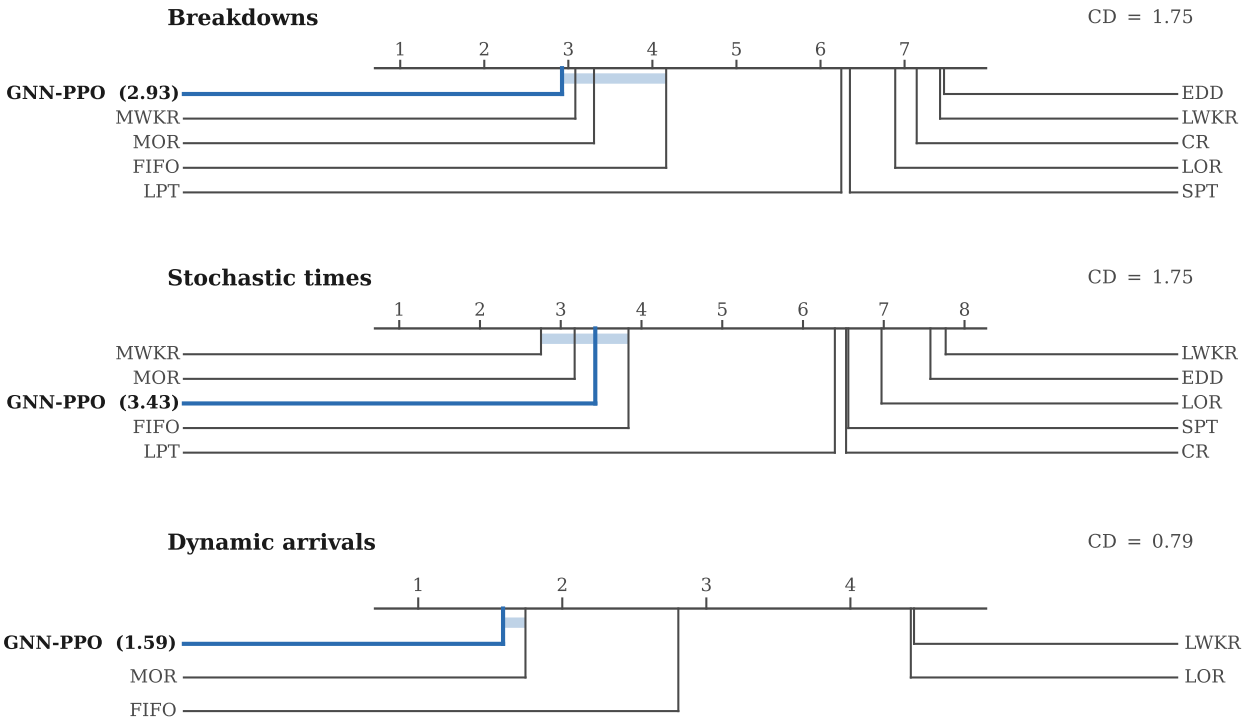


Figure 4: Critical-difference diagrams on realised quality, one per disruption class. Methods joined by the shaded bar are not distinguishable from the best-ranked method at $\alpha = 0.05$.

Table 5
The same policy against three definitions of "best dispatching rule". Deviations are relative percentage deviation from the certified optimum; the difference column is the median paired difference in percentage points, with Wilcoxon z, p and rank-biserial effect size over the paired instances; $N = 43$ for the learned-policy rows and $N = 40$ where exact ties are dropped, as the n column records.

Comparator	Its RPD	Method	Its RPD	Median diff.	n	z	p	r_{rb}	Verdict
$\mathrm{PDR}_{\mathrm{global}}$	251.4	**GNN–PPO**	24.8	+233.4	43	-5.71	1.1×10^{-8}	+1.000	WINS
	251.4	GA	5.5	+251.5	43	-5.71	1.1×10^{-8}	+1.000	WINS
	251.4	CP-SAT	0.0	+254.6	43	-5.71	1.1×10^{-8}	+1.000	WINS
$\mathrm{PDR}_{\mathrm{family}}$	148.1	**GNN–PPO**	24.8	+132.4	43	-5.71	1.1×10^{-8}	+1.000	WINS
	148.1	GA	5.5	+144.3	43	-5.71	1.1×10^{-8}	+1.000	WINS
	148.1	CP-SAT	0.0	+152.1	43	-5.71	1.1×10^{-8}	+1.000	WINS
$\mathrm{PDR}_{\mathrm{oracle}}$	19.0	**GNN–PPO**	24.8	−7.6	43	-4.11	4.0×10^{-5}	−0.719	LOSES
	19.0	GA	5.5	+11.5	40	-5.51	3.6×10^{-8}	+1.000	WINS
	19.0	CP-SAT	0.0	+21.1	40	-5.51	3.6×10^{-8}	+1.000	WINS

learned policy against 220.1% for the critical-ratio rule, and 24.8% against 430.3% over the full 43-instance nominal set — the condition is not a corner case.

The remedy is cheap. Realised quality is computable from records that any such study already possesses, provided a certified optimum or an accepted bound is available for the nominal instance; it requires no re-execution. We would encourage reporting r, q and $q - r$ together, and reserving robustness language for the last of the three.

7.3. A reporting protocol

The audit suggests four requirements, none onerous.

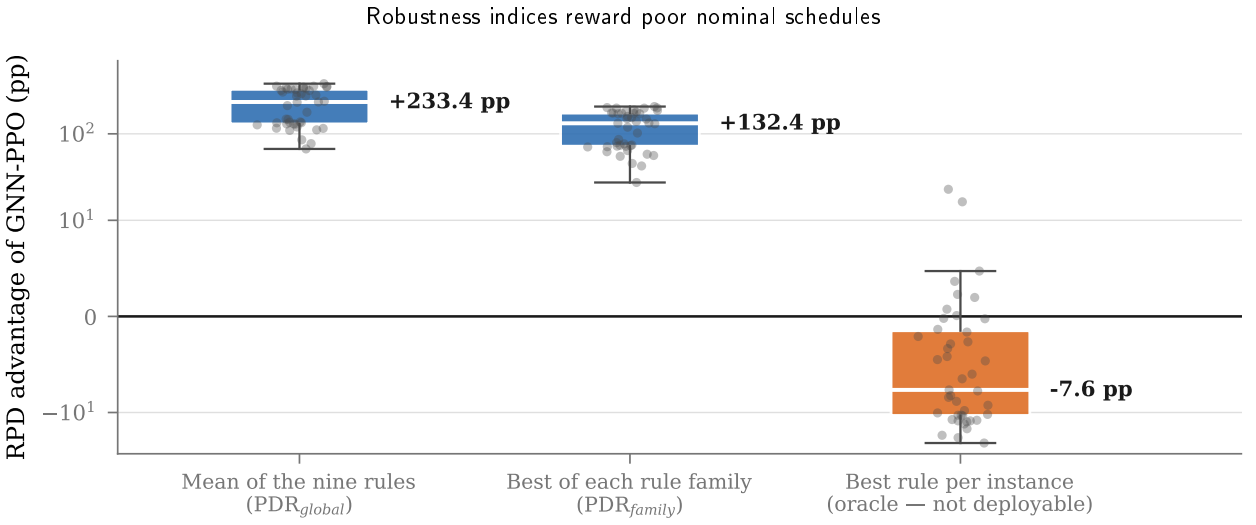


Figure 5: Advantage of the learned policy over three definitions of the dispatching-rule baseline, per instance. The vertical scale is symmetric-log. The oracle comparator is included for calibration and is not deployable.

(1) Grade disrupted performance against a reference external to the method: a certified optimum where one exists, otherwise a fixed bound shared by all methods and named as such.
(2) Report nominal deviation alongside any robustness quantity, so that readers can perform the decomposition of Proposition 1 themselves.
(3) Report a tail statistic, not only a mean.
(4) Name the comparator. If the claim is against "the best dispatching rule", state which of the three constructions of Section 4.4 is meant, and prefer reporting more than one.

7.4. Threats to validity

We state the limits of this study plainly, because several of them bound the conclusions materially.

Instance breadth in the disruption grid. The 9 regimes are executed on two instances, `ft06` and `la01`. The analytic result of Proposition 1 is instance-independent, and the empirical correlations are computed across methods rather than across instances, so the audit's central claim does not rest on instance breadth. The method-level findings of Section 6.4 do, and should be read as holding for these two instances and this disruption model rather than as a general ranking.

Equivalence from non-rejection. A Nemenyi non-rejection is a failure to distinguish, not a demonstration of equivalence. With $k = 10$ and $N = 60$ the critical difference is 1.75 on a rank scale of 1 to 10, which is wide; the statement that the policy is "not significantly worse" should be read with that width in mind rather than as a claim of parity.

Training seeds. The learned policy is a single trained checkpoint. Its position in Table 4 therefore carries no estimate of training variance, and the reinforcement-learning reproducibility literature is emphatic that this variance can be large (Henderson et al., 2018; Agarwal et al., 2021). Multi-seed training infrastructure is implemented and reported in the accompanying repository, but the multi-seed campaign had not completed at the time of writing and no multi-seed number is claimed here.

Method coverage under disruption. The nine-regime grid compares the learned policy against nine dispatching rules in six regimes and four in the other three (Section 3.3). CP-SAT and the genetic algorithm enter only through the descriptive comparison of Section 6.6, at one instance, one regime and five seeds. Consequently, wherever this paper says the policy is not significantly worse than the best method, the comparison set is the dispatching rules in that regime, not the exact and metaheuristic solvers — and in the single regime where those solvers were run they finish ahead of the policy (42.0% and 41.3% against

48.0%), though not significantly so at $n = 4$. A full offline-method arm across the grid is the natural next experiment and would sharpen Section 6.6 from an observation into a test.

Rule coverage under arrivals, and why it is not conservative. Five of the ten methods are compared in the arrival regimes, for the reasons given in Section 3.3. This cuts in two directions and we want to be explicit about both. For the measurement claim it is close to neutral: the excluded set spans the range: it contains the two rules with the worst nominal schedules (CR, EDD) and also the rule with the second best (MWKR, at 30.7%), so it does not manufacture the negative correlation — which in any case rests on the 6 ten-method regimes. For the claim about the learned policy it is *anti*-conservative: MWKR, which outranks the policy in five of the six non-arrival regimes, is excluded, so the policy's first rank under arrivals is a first rank in a weakened field. We therefore do not treat the arrival result as the strongest evidence for the policy, and a reader who discounts it entirely is left with the measurement argument, which does not depend on it.

Reference validity under arrivals. As noted in Section 4, RQ under dynamic arrivals is referenced to the optimum of the nominal instance rather than of the realised one. Comparisons within a regime remain valid; absolute magnitudes there are not optimality gaps.

Post-hoc status of the re-measurement. The measurement audit was not pre-registered. It was prompted by an examination of the metric and is reported as post-hoc analysis of a frozen corpus. No datum was altered, and the pre-registered verdicts computed on the original metric stand as recorded in the accompanying materials.

8. Conclusion

The robustness index that dominates evaluation of dynamic scheduling under disruption normalises a method's degradation by its own nominal makespan. We proved that this makes the index strictly decreasing in nominal deviation at fixed realised performance, so that the worse a method's plan, the more robust it appears, and that a negative index — widely read as exceptional resilience — is the signature of a plan poor enough for perturbation to improve it. In a pre-registered corpus of 4,500 disrupted executions across 9 regimes, the index is negatively rank-correlated with nominal deviation everywhere and at $\rho_s \in [-0.985, -0.948]$ in the 6 regimes with a ten-method field, it orders methods opposite to realised quality in every regime, and following it costs up to 49.3 percentage points of deviation from the optimum.

Realised quality — the deviation of the executed makespan from a certified optimum — together with a CVaR_{90} tail and an explicitly named dispatching comparator, repairs the comparison at the cost of re-measurement alone. Re-measured this way, the heterogeneous GNN–PPO policy is not significantly worse than the best-ranked dispatching rule in any of the 9 regimes and takes first rank in 5, including all three dynamic-arrival regimes, where the value of deciding online is greatest — in every case tied with 1 rule under arrivals and 3 elsewhere, rather than separated from the field. The conventional index places the same policy near the bottom of the same runs.

The wider point is methodological. A metric whose denominator is chosen by the method being graded cannot support a comparison between methods, and the scheduling literature has a well-established external reference available in the certified optima of its standard benchmarks. Using it costs nothing and settles the question the index was introduced to answer.

CRediT authorship contribution statement

Joseph Javier Sánchez Acuña: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

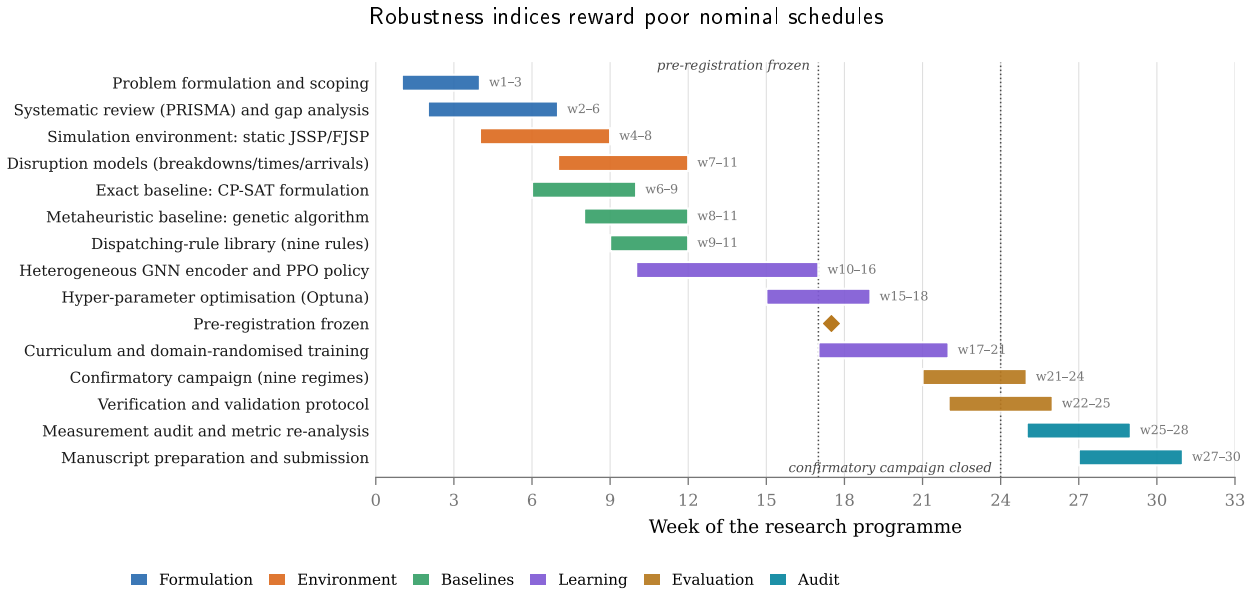


Figure 6: Schedule of the research programme. The diamond marks the pre-registration freeze, after which no hypothesis, threshold, instance set or statistical test was altered. The measurement audit and the re-analysis reported here occupy the final phase and are post-hoc with respect to that freeze.

Data availability

The source code, the pre-registration document, the raw per-run records underlying every table and figure, and the analysis scripts that regenerate them are available in the accompanying public repository. Every number reported in Section 6, including the identity audit of Section 4 and the offline-method comparison of Section 6.6, is produced by those scripts from those records.

A. Research programme

Figure 6 sets out the schedule of the research programme from which the corpus arises, including the point at which the pre-registration was frozen and the point at which the confirmatory campaign closed. The measurement audit reported in this paper follows both.

CRediT authorship contribution statement

Joseph Javier Sánchez Acuña: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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